- Instead of partitioning points using splitting line passing through dataset's points, we can divide a region in two balanced halves with respect to the number of points or the sub-regions size;
- Instead of cycling through dimensions, we can choose at every step the dimension with the greatest spread or variance, and store the choice made in each tree node.
- Instead of storing points in nodes, each node could describe a region of space and link (directly or indirectly) to an array containing the actual elements.

These solutions are the basis of the data structures we will discuss in this chapter, *R-trees* and *SS-trees*.

10.2 R-tree

The first evolution of k-d trees we will present are R-trees. Although we won't delve into the details of their implementation, we are going to discuss the idea behind this solution, why it works, and their high-level mechanism.

R-trees were introduced in 1984 by Antonin Guttman in the paper R-TREES. A DYNAMIC INDEX STRUCTURE FOR SPATIAL SEARCHING.

They are inspired by B- trees[2], balanced trees with a hierarchical structure; in particular Guttman used as a starting point B+ trees, a variant where only leaf nodes contain data, while inner nodes only contain keys and serve the purpose to hierarchically partition data.